

Computational Structures and the Quantum Information Architecture of Spacetime

Research Statement

An empirical investigation of numerical methods that spontaneously generate mathematical structures matching fundamental cosmological parameters, revealing connections between computational complexity, information theory, and physical law.

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1 Introduction

The quantum information structure of spacetime represents one of the most promising frameworks for unifying gravitational physics, quantum field theory, and information theory. This perspective posits that the geometric and topological properties we observe in spacetime may not be fundamental but rather emergent from deeper computational and information-theoretic substrates. From holographic principles in string theory to tensor network approaches in condensed matter physics, evidence accumulates that spacetime geometry encodes—or is encoded by—patterns of quantum entanglement and information flow.

Empirical Observation

A data-centric framework developed for biomass gasification systems employs elliptic integrals to achieve mass balance closure. The gains returned by this method consistently fall in the 10^{-47} to 10^{-60} range—the numerical territory occupied by the cosmological constant in Einstein’s field equations.

This observation raises fundamental questions: When numerical methods applied to complex physical systems spontaneously generate mathematical objects whose characteristic scales match fundamental cosmological parameters, what does this reveal about the relationship between computational structures and physical theories? How do information-preserving embeddings, algorithmic complexity measures, and topological invariants—all central to quantum information approaches—manifest in classical non-equilibrium systems? Can data-driven methods illuminate connections between conservation laws and geometric structures that traditional approaches obscure?

This statement presents the methodological framework, empirical evidence, structural diagnostics, and falsifiable predictions that make this line of investigation tractable. The aim is neither to claim immediate resolution of fundamental questions nor to propose speculative connections, but rather to establish a rigorous empirical basis for systematic inquiry into how computational methods for multi-physics systems may encode information about deeper organizational principles in nature.

2 The Physical System and Computational Challenge

Fluidized bed gasification of biomass converts solid carbonaceous material into synthetic gas through thermochemical reactions in a turbulent multiphase environment. The system presents considerable modeling difficulty: fuel heterogeneity (varying moisture, ash, and volatile content), non-equilibrium kinetics (competing reaction pathways with temperature-dependent rates), and coupled transport phenomena (simultaneous mass, momentum, and energy exchange across gas-solid interfaces) create a problem that resists first-principles characterization.

Traditional computational approaches solve coupled systems of partial differential equations describing conservation of mass, momentum, energy, and chemical species. These require closure models for turbulent mixing, uncertain boundary conditions at moving particle surfaces, and empirical correlations for heat and mass transfer coefficients. The resulting computations are intensive, and

model uncertainties compound through the coupling structure.

The framework under investigation takes an alternative path: rather than solving conservation equations directly, it constructs a representation that achieves the same outcome—mass closure in the fluid field—from experimental data through a sequence of transformations culminating in an elliptic integral evaluation. This approach is constructive in a specific sense: it builds the same closure object (a transfer function with physically admissible poles, zeros, and gain) that conservation-based methods target, but does so from observed system behavior rather than postulated governing equations.

3 Methodological Framework

3.1 The three-argument structure

The elliptic integral framework employs three quantities as arguments:

- **Temperature (T):** Absolute temperature serves as the primary thermodynamic state variable. Its inclusion requires no special justification—it is the most fundamental descriptor of thermal state and enters naturally into any energy-based formulation.
- **Specific computational time (sct):** This metric emerges from matrix shearing operations applied to the computational representation of the system. It quantifies algorithmic stiffness—the resistance of the numerical problem to iterative solution. In reactive flow systems, stiffness correlates strongly with thermodynamic non-equilibrium: conditions far from equilibrium generate numerically stiff problems due to disparate timescales in the coupled rate equations. The sct operationalizes this connection, providing a data-derived proxy for the degree of disequilibrium.

The use of computational time as a physical diagnostic is uncommon. Most applications treat algorithmic cost as a practical constraint divorced from system physics. Here, it functions as an independent coordinate capturing aspects of the process state that temperature alone cannot describe. This suggests a deeper relationship between information-theoretic resources (algorithmic complexity) and physical state spaces—a connection central to quantum information approaches to fundamental physics.

- **Composite parameter (κ):** This quantity consolidates diverse factors influencing the process: biomass composition (cellulose, hemicellulose, lignin fractions), particle characteristics (size distribution, density), and operating conditions (fluidization velocity, bed geometry, steam-to-fuel ratio). These variables vary across orders of magnitude and lack natural commensurability.

The composite parameter construction employs canonical correlation analysis to identify stable patterns of co-variation, geometric projections to establish dimensional compatibility, and neural-statistical synthesis to preserve case identity while achieving scale equitability. The result is an information-conserving embedding: a map from high-variance heterogeneous inputs to a common representational space where integration becomes meaningful.

This construction respects a fundamental principle: an integral represents a sum, and meaningful summation requires entities that inhabit compatible spaces. The embedding ensures this compatibility not through arbitrary normalization but through structure-preserving transformation that maintains distinguishability while establishing commensurability.

3.2 From elliptic integrals to transfer functions

The complete elliptic integral of the first kind, implemented via the arithmetic-geometric mean (AGM) algorithm, accepts the triplet (T, sct, κ) and returns a linear time-invariant system:

$$H(s) = k \frac{\prod_{i=1}^{n_z} (s - z_i)}{\prod_{j=1}^{n_p} (s - p_j)}, \quad (1)$$

where z_i are zeros (numerator roots), p_j are poles (denominator roots), and $k > 0$ is the gain.

The use of three-argument elliptic integrals has precedent in celestial mechanics, where they solve orbital dynamics problems. Their application to energy systems represents a transfer of mathematically mature tools into a new physical domain—not model whimsy but a deliberate exploitation of known convergence and stability properties.

Table 1: Transfer function realizations from four representative cases

	Case 1	Case 2	Case 3	Case 4	
Zeros	14×1 complex	16×1 complex	16×1 complex	18×1 complex	<i>Spanning</i>
Poles	15×1 complex	16×1 complex	17×1 complex	18×1 complex	
Gain k	1.45×10^{-47}	3.98×10^{-53}	1.92×10^{-58}	1.26×10^{-60}	

hardwood, agricultural residue, and energy crop feedstocks at varying temperatures (700-850°C) and steam ratios (0.3-0.7). System orders vary modestly while gains span thirteen orders of magnitude.

The zeros and poles emerge as complex conjugate pairs satisfying standard stability criteria for causal systems. The gains vary over thirteen orders of magnitude yet consistently occupy the 10^{-47} to 10^{-60} band. Both features—the constrained pole-zero structure and the systematic gain magnitude—prove consequential for establishing empirical correlations with process observables.

4 The Scale Alignment and Its Implications

4.1 Connection to general relativity

In general relativity, fluid solutions are exact solutions to Einstein’s field equation:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}, \quad (2)$$

where Λ is the cosmological constant representing vacuum energy density. The term $\Lambda g_{\mu\nu}$ contributes a spatially uniform gravitational effect throughout spacetime.

The gains computed by the elliptic integral framework occupy the same numerical band as Λ

values under appropriate unit normalization. This is not claimed as a universal identity independent of dimensional analysis, but rather as an operational correspondence: there exists a unit-consistent normalization under which k maps to the role Λ plays in the field equation.

What prevents this from being dismissed as numerology is the dual functionality of k . The same constant that achieves mass balance closure in the thermochemical system also anchors correlations with experimental performance metrics. It serves simultaneously as a computational output necessary for conservation and as an empirical predictor of system behavior. This dual role suggests the quantity captures something structural rather than serving as a free-floating fit parameter.

4.2 Operational bridge construction

Define $\Lambda_S \equiv k$ for each sample and Λ_G as the geometric mean across the ensemble. Construct the dimensionless log-deviation:

$$\xi \equiv \ln \left(\frac{\Lambda_S}{\Lambda_G} \right). \quad (3)$$

This statistic is two-tailed: $\xi > 0$ indicates above-mean behavior, $\xi < 0$ below-mean. Local variations then take the form

$$\Lambda(x) = \Lambda_G e^{\xi(x)}, \quad (4)$$

providing a map from experimental fluid behavior to a general-relativistic source term structure. The formulation captures spatial, temporal, or operating-point variations in the constitutive constant without requiring the absolute scale to match cosmological measurements—only that the relative structure and the functional form correspond.

5 Empirical Diagnostics and Verification

Two quantities extracted from the transfer function realizations exhibit systematic relationships with process performance:

5.1 Continuous diagnostic: The ξ statistic

Product gas yield demonstrates monotonic dependence on $\xi = \ln(\Lambda_S/\Lambda_G)$ across operating regimes. This relationship emerges at the bin level—aggregating over sufficiently many cases to average measurement noise while preserving trends. Higher yields correspond to systematic deviations in one direction from the geometric mean; lower yields to deviations in the other.

The correlation is not fitted. The ξ statistic derives from the gain magnitude, which itself emerges from the elliptic integral evaluation constrained to achieve mass closure. That this same quantity independently tracks experimental yield provides empirical anchor for the framework's physical validity.

5.2 Discrete diagnostic: The Squarity index

Define the Squarity index as $S = \mathbb{K}[n_z \neq n_p]$, indicating whether pole and zero counts differ. This binary indicator correlates with carbon conversion efficiency and exhibits "locking" behavior at high conversion levels: S stabilizes and ceases to flip between values under small perturbations.

The locking suggests a topological transition in the space of admissible transfer function realizations. At high conversion, the system enters a regime where the structural invariant (equal or unequal pole-zero counts) becomes insensitive to operational variations—analogue to topological phases in condensed matter systems where discrete invariants remain stable under continuous deformations.

5.3 The role of computational time

As operating conditions smooth—better mixing, longer residence times, narrower temperature gradients—the set metric decreases and the distribution of ξ collapses toward zero. Mass closure accuracy improves superlinearly in this regime, consistent with known acceleration of AGM convergence near low eccentricity in elliptic integral evaluation.

This behavior suggests eccentricity in the mathematical formulation serves as a geometric measure of thermodynamic non-equilibrium. Systems far from ideal behavior (large temperature gradients, incomplete reactions, phase segregation) correspond to high eccentricity; systems approaching equilibrium correspond to low eccentricity with attendant convergence acceleration.

6 Convergence to Perfect-Fluid Limits

Define a composite non-equilibrium index:

$$\epsilon \equiv \text{median}(|\xi|) + \alpha \text{Var}(\text{sct}) + \beta \mathcal{K}[S = 1], \quad (5)$$

with weights $\alpha, \beta > 0$. As $\epsilon \rightarrow 0$ corresponding to equilibrium conditions, smooth fields, and saturated pole-zero interlacing—the following properties converge:

- Mass closure residuals vanish
- The ξ distribution collapses (median and variance approach zero)
- The Squarity index locks at $S = 0$ (equal pole and zero counts)
- The set metric contracts toward minimal values

In this limit, the approximate solution from the data-centric framework coincides with exact solutions for perfect fluids—Euler fluids in classical mechanics or the stress-energy tensor for perfect fluids in general relativity with constant Λ . The convergence is structural: the transfer function degenerates to a form consistent with idealized physics where dissipation vanishes and local equilibrium holds.

This establishes a theoretical pathway connecting data-derived representations to limiting cases grounded in fundamental theory. It provides operational meaning to "approaching equilibrium"—not

a vague qualitative statement but a measurable trajectory in (ξ, sct, S) space with quantifiable convergence rates.

7 Connections to Quantum Information Frameworks

Three aspects of this work connect directly to questions central in quantum information approaches to spacetime:

7.1 Information-preserving embeddings

The composite parameter construction employs canonical correlation and geometric regularization to create low-distortion maps from high-dimensional heterogeneous data to lower-dimensional manifolds. This parallels proposals where spacetime geometry emerges from entanglement patterns in quantum states. Both scenarios require transformations that compress dimension while maintaining distinguishability and preserving structural relationships.

In quantum gravity approaches, the question is how bulk spacetime geometry emerges from boundary quantum information. Here, the question is how process-level complexity (dozens of coupled variables with disparate scales) maps into a single parameter that meaningfully enters an integral formulation. Success in the thermochemical context demonstrates that information-conserving dimensional reduction can work in classical non-equilibrium settings, suggesting the technique may be more broadly applicable than typically assumed.

7.2 Computational resources as physical coordinates

The sct metric establishes a direct link between algorithmic complexity (iterative solution cost) and physical state (thermodynamic non-equilibrium). Such connections remain uncommon. The result here—that computational stiffness serves as a legitimate thermodynamic coordinate entering fundamental equations alongside temperature—strengthens the case that information-theoretic resources and physical state spaces share deep connections.

In quantum information approaches to gravity, computational complexity of state preparation or dynamics simulation plays central roles. Circuit complexity measures distance in the space of quantum states; algorithmic cooling protocols manipulate entropy; quantum error correction relates to bulk-boundary correspondence. The patterns observed here suggest analogous structures may operate in classical non-equilibrium thermodynamics, with algorithmic stiffness serving as a diagnostic for departure from idealized limiting behavior.

7.3 Topological invariants and discrete diagnostics

The Squarity index exhibits discrete transitions and locking behavior reminiscent of topological phases. Topological invariants provide robust diagnostics precisely because they count global features rather than measure local properties—they remain stable under continuous deformations. Their appearance in thermochemical conversion dynamics via purely data-driven methods suggests

topological characterization may be more general than currently recognized.

In proposed quantum gravity frameworks, topology changes in spacetime manifolds play fundamental roles (baby universe formation, black hole evaporation, wormhole physics). Observing topology-like transitions in engineered systems through data-driven methods opens possibilities for studying such phenomena in experimentally accessible contexts where controlled perturbations and systematic measurements become feasible.

8 Falsifiable Predictions and Experimental Tests

The framework generates specific expectations amenable to controlled testing:

- **Cross-process portability:** Applying the identical pipeline to pyrolysis, torrefaction, or combustion should preserve three features: (i) mass closure via $H(s)$ of the same structural form, (ii) monotonic ξ -yield correlation, and (iii) S functioning as a discrete efficiency marker. Only the geometric mean Λ_G should require recalibration. Failure would indicate gasification-specific results rather than general principles.
- **Ablation tests:** Removing sct from the integral arguments should degrade mass closure and break ξ -yield monotonicity. Replacing canonical correlation with random embedding should destroy case identity preservation. Eliminating geometric regularization should cause the composite parameter to lose information-conserving properties. Each ablation targets a specific structural claim with predicted failure modes.
- **Convergence quantification:** Constructing operating sequences that smoothly approach equilibrium should produce measurable trajectories in (ξ, sct, S) space. Mass closure error should decrease superlinearly as conditions smooth (AGM acceleration). The Squarity index should lock at high conversion. The sct metric should contract. Quantifying these rates tests the convergence framework.
- **Field reconstruction:** Mapping $\Lambda(x) = \Lambda_G e^{\xi(x)}$ over reactor spatial coordinates should produce smooth fields with physically plausible gradients. Spurious oscillations or unphysical discontinuities would indicate ξ captures statistical noise rather than genuine spatial structure.
- **Noise robustness:** Moderate observational noise should perturb individual zeros and poles while leaving ξ (a ratio to geometric mean) statistically stable. Yield predictions should degrade gracefully rather than collapsing catastrophically, demonstrating the logarithmic formulation provides natural regularization.

9 Research Trajectory

The immediate research trajectory follows directly from the falsification framework outlined above. Testing cross-process portability requires assembling pyrolysis and torrefaction datasets with comparable measurement quality and applying the pipeline without modification. Quantifying

ablation effects demands systematic component removal with careful measurement of degradation in mass closure, correlation strength, and diagnostic stability. Measuring convergence rates requires designing operating sequences that traverse from high to low non-equilibrium while maintaining measurement consistency.

Beyond immediate validation, three deeper questions structure longer-term investigation:

1. **Eccentricity as disequilibrium measure:** Formalizing the connection between elliptic integral eccentricity and thermodynamic non-equilibrium distance. Does eccentricity provide a universal metric for departure from perfect-fluid behavior? Can it be related to standard measures of irreversibility (entropy production, exergy destruction, Carnot efficiency)?
2. **Information geometry:** Does the composite parameter construction define a natural Riemannian metric on the space of operating conditions? If so, do geodesics in this space correspond to optimal process trajectories (minimal dissipation, maximum conversion, fastest equilibration)? Can variational principles be formulated where physical processes extremize action functionals defined on the embedded manifold?
3. **Universal computational structures:** What minimal requirements must a multi-physics problem satisfy for elliptic integral closure to succeed? Is it conservation law structure, scale separation, or something more subtle about how information flows through coupled field equations? Characterizing these requirements would indicate whether the approach addresses a broad class of problems or remains specific to reactive flows.

10 Conclusion

This research statement presents an empirical observation that merits systematic investigation: a data-centric framework for complex thermochemical processes spontaneously generates transfer function gains matching cosmological constant scales. This alignment is accompanied by structural diagnostics that independently correlate with experimental observables and convergence properties that connect to perfect-fluid limits in fundamental theory.

The observation raises questions about relationships between computational structures and physical theories. When numerical methods generate mathematical objects whose scales match fundamental parameters, when algorithmic complexity proxies physical state, when topological invariants diagnose phase-like transitions—these patterns suggest connections between information-theoretic frameworks and physical law that merit rigorous investigation.

The quantum information structure of spacetime provides a natural context for this investigation. Proposals in that domain emphasize emergence of geometry from information, computational complexity as physical observable, and topological invariants as robust diagnostics. The patterns observed here in classical non-equilibrium thermodynamics exhibit striking parallels, suggesting these may be general features of how mathematical structure relates to physical description rather than artifacts specific to quantum gravity.

Whether these connections ultimately illuminate fundamental questions about spacetime architecture or remain useful but domain-specific techniques will be determined through systematic empirical investigation. The framework is falsifiable, the predictions are specific, and the experimental tests are tractable. The next steps are clear: test portability, quantify ablations, measure convergence, reconstruct fields. The observations are documented, the implications are constrained by appropriate epistemic humility, and the path forward follows directly from established scientific method.